Randomness expansion from untrusted quantum devices

Carl A. Miller

University of Michigan, Ann Arbor

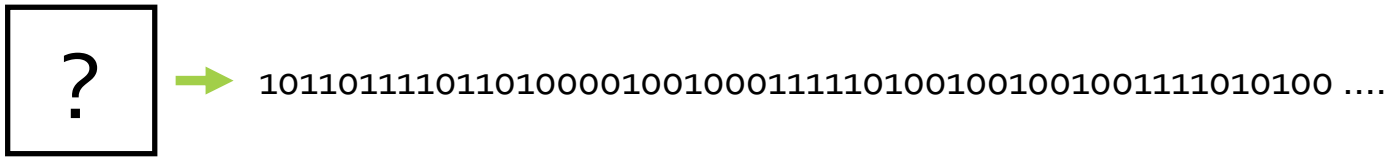
Based on "Universal security for randomness expansion from the spot-checking protocol" (arXiv:1411.6608), with Yaoyun Shi.





The question

Can we generate <u>provable</u> random numbers?



Why it matters

Security of protocols like RSA breaks down if randomness is bad.
[Lenstra+ 12, Heninger+ 12]



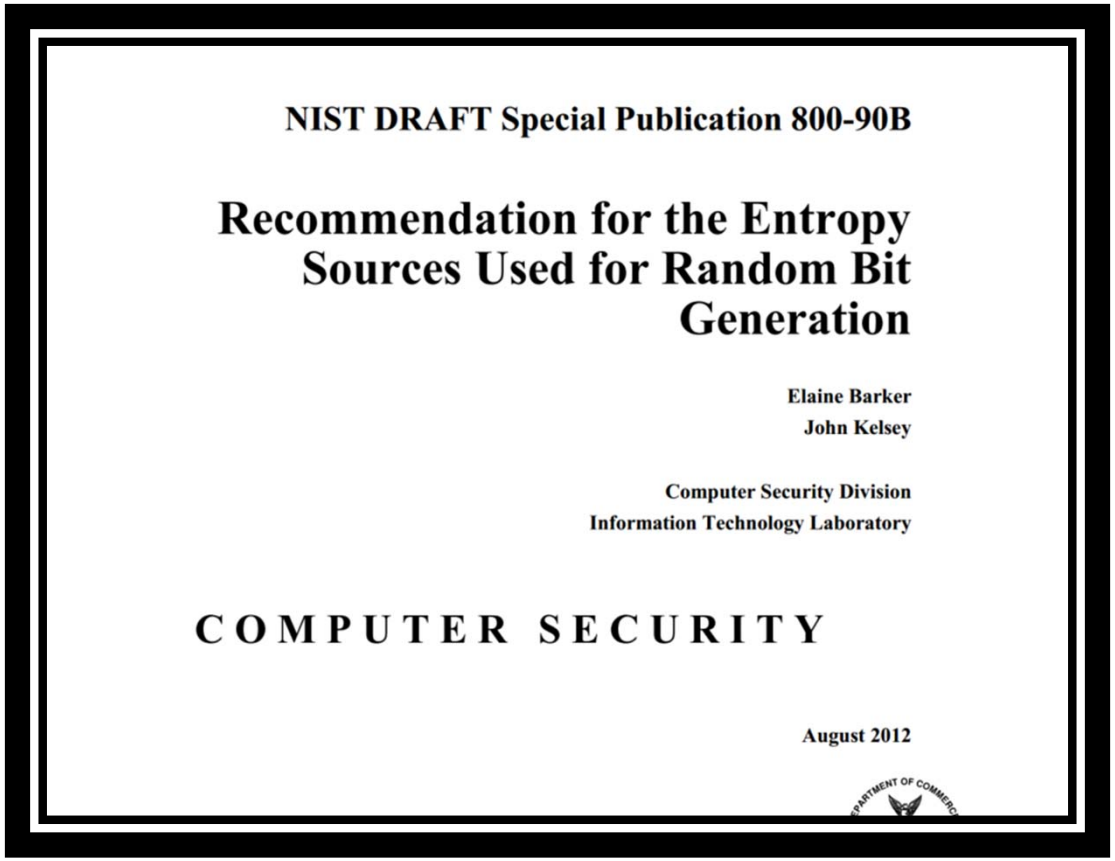
P,Q (primes)

P,Q





Existing solutions



NIST DRAFT Special Publication 800-90B

Recommendation for the Entropy Sources Used for Random Bit Generation

Elaine Barker
John Kelsey

Computer Security Division
Information Technology Laboratory

COMPUTER SECURITY

August 2012

"[We assume] that the developer understands the behavior of the entropy source and has made a **good-faith effort** to produce a consistent source of entropy."

Can we generate randomness without assuming good faith?

Quantum random number generation

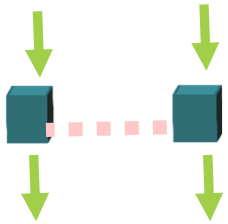
- Untrusted-device randomness expansion
- Untrusted-device randomness amplification
- Semi-device-independent random number generation.
- Contextuality-based randomness expansion.
- Randomness extraction.

Quantum random number generation

- Untrusted-device randomness expansion

Small uniform seed + untrusted device -> uniform randomness

0 0 1 1 1 0 1 1



1010111101100010011011001111101100110111101111111110100001010001001111110101010111010101010 ....

Only assumption: Non-communication.

Results

Timeline

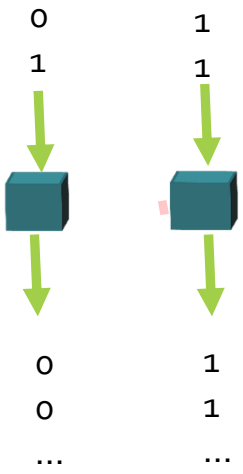
Colbeck

Us

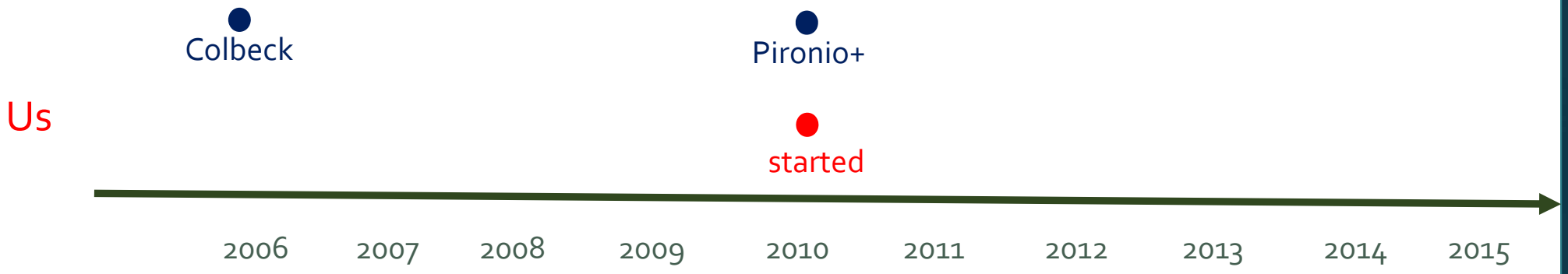
2006 2007 2008 2009 2010 2011 2012 2013 2014 2015

Colbeck proposed a protocol based on **repeating a nonlocal game.**





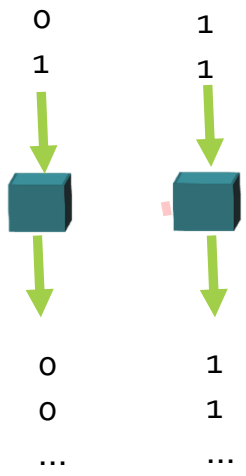
Timeline



Pironio+: analysis & experiment.

I started working in quantum information.

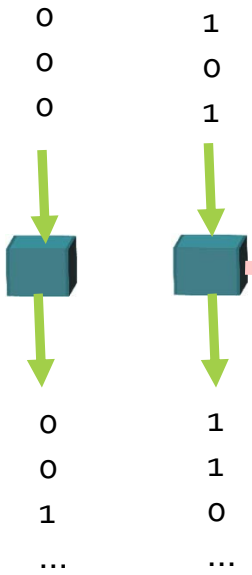




The challenge of the entangled adversary

Quantum information can be **locked** – accessible *only* to entangled adversaries. [E.g., DiVincenzo+ 04]







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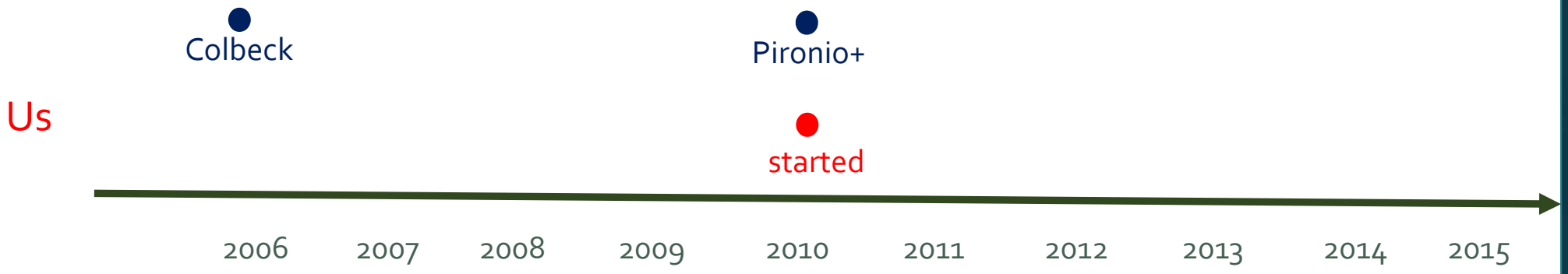
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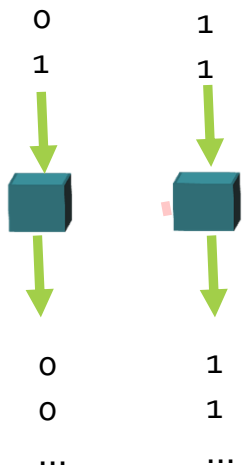
Timeline



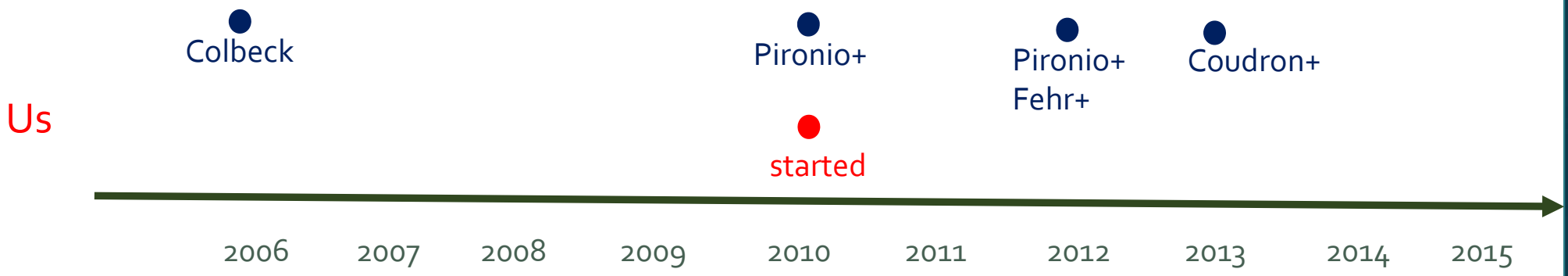
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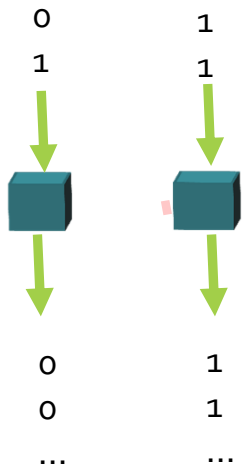
Timeline



Pironio+, Fehr+, Coudron+ proved security against unentangled adversary.

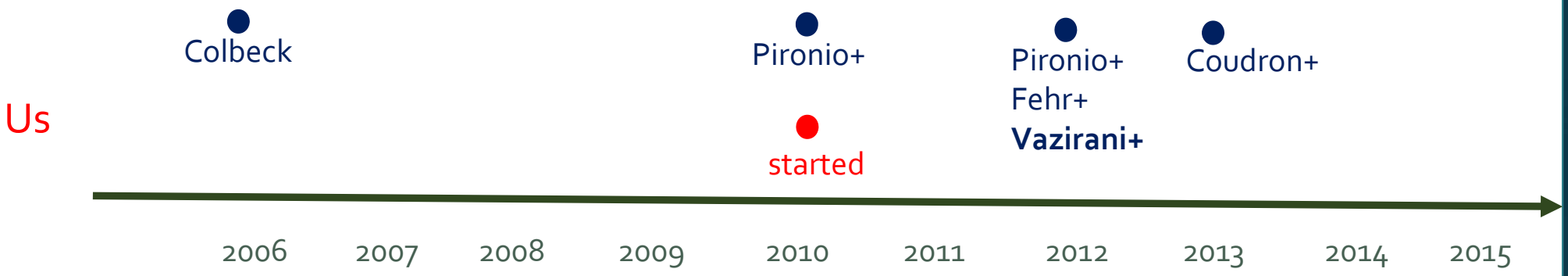
Methods: Classical statistical arguments (e.g., Azuma's inequality).







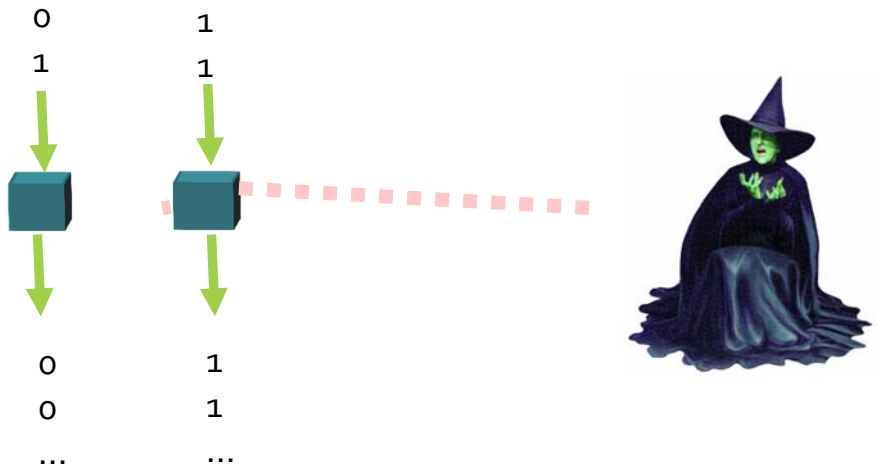
Timeline



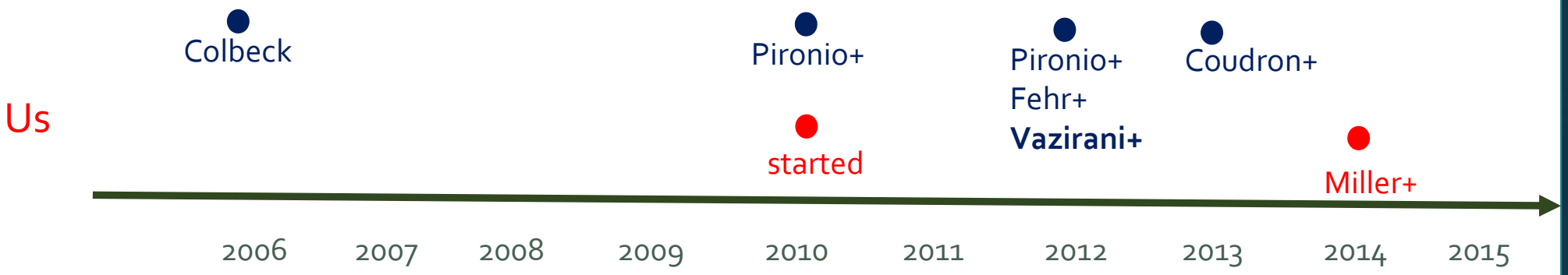
Vazirani-Vidick proved full security against an entangled adversary (no error tolerance).

Method: Insecurity implies communication paradox.





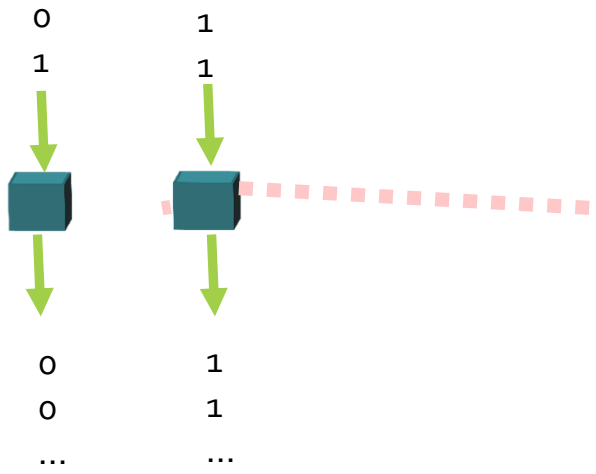
Timeline



We proved full security, with error-tolerance >1.4%.

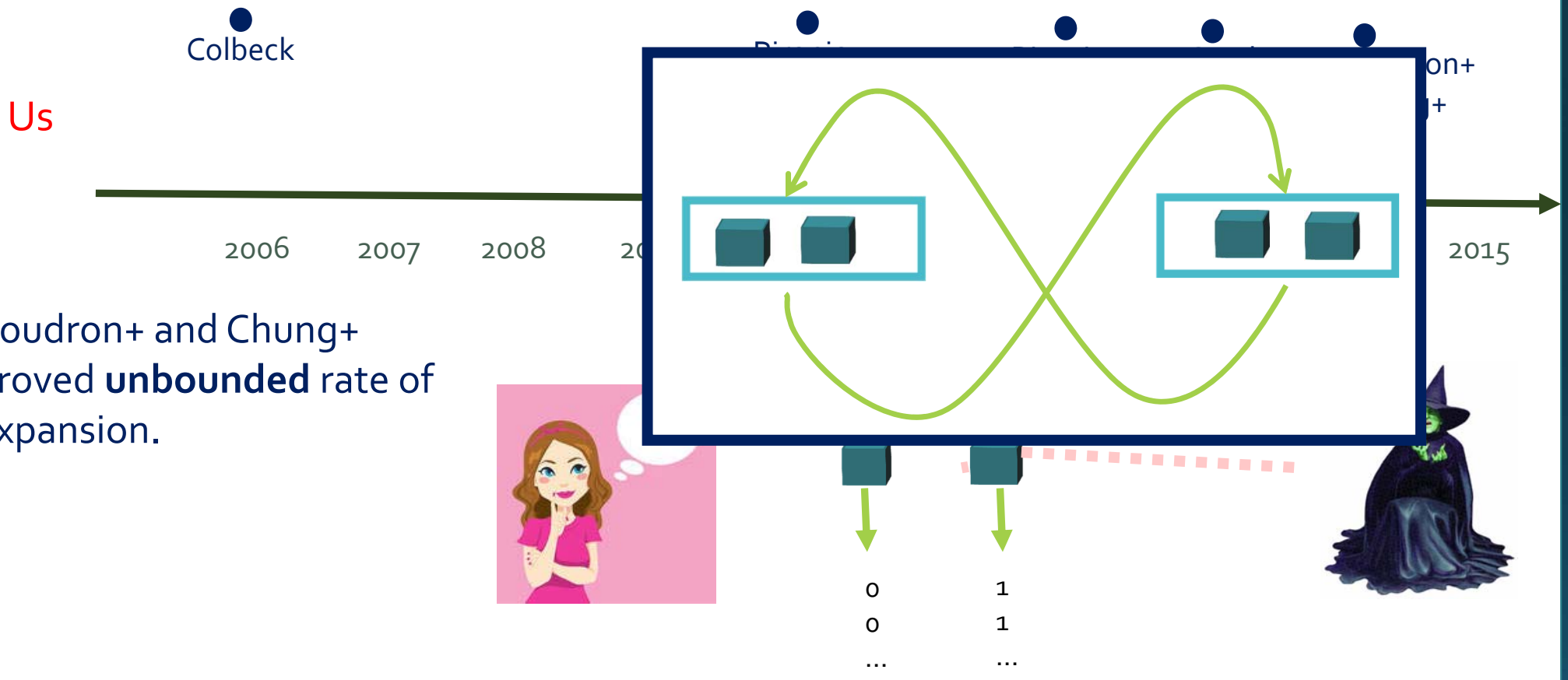
Method: Constraints on
$X \mapsto Tr[X^{1+\varepsilon}]$
for adversary-output states.





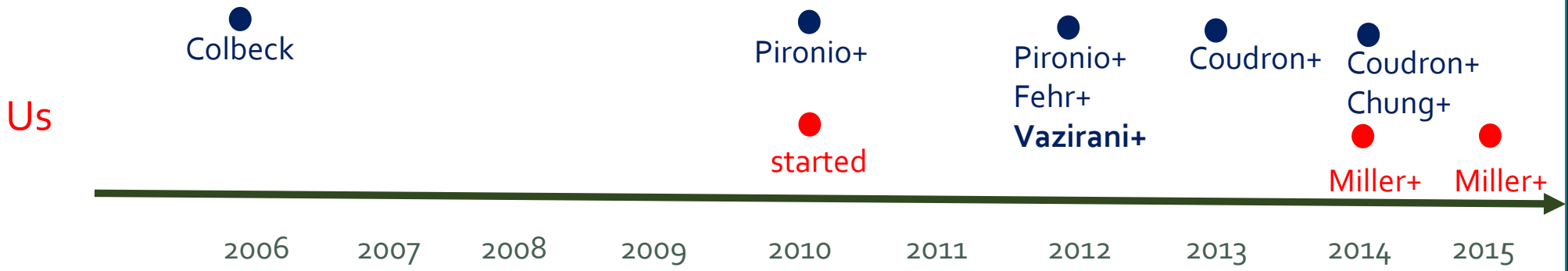


Timeline



Coudron+ and Chung+ proved **unbounded** rate of expansion.

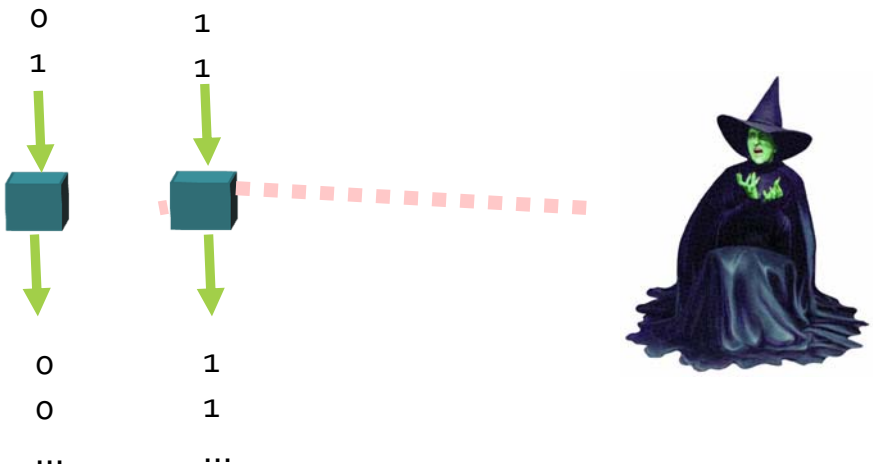
Timeline



Coudron+ and Chung+ proved **unbounded** rate of expansion.

Our current paper: Covers all nonlocal games, maximal noise tolerance.





Current Result

The spot-checking protocol

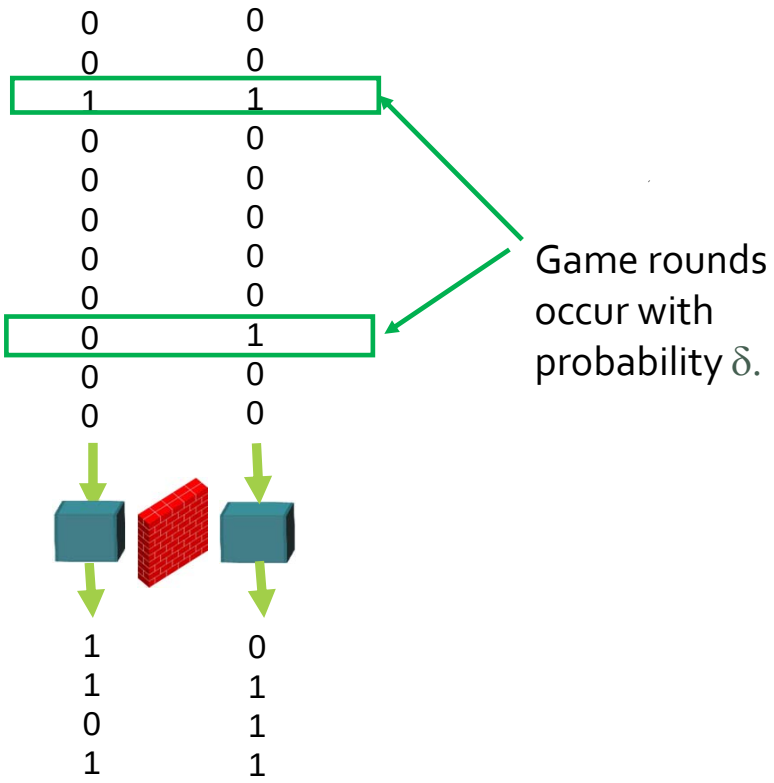
(Coudron-Vidick-Yuen 2013, Vazirani-Vidick 2012)

Let G = nonlocal game, a = fixed input.

1. Run the device N times. During "game rounds," play G. Otherwise, just input a.

2. If the average score during game rounds was < C, abort.

3. Otherwise, apply randomness extractor.



The spot-checking protoco

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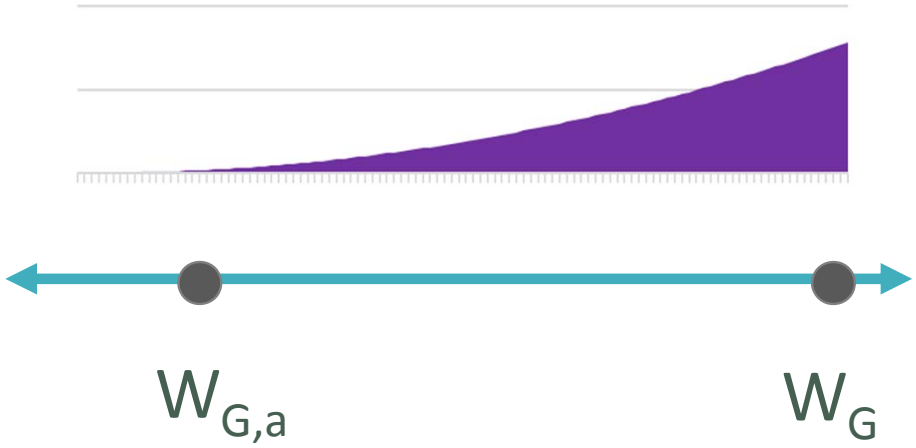
Let $W_{G,a}$ = optimal score among devices that are <u>deterministic on a.</u>

Thm (CVY 13): The protocol is secure against an unentangled adversary if $C > W_{G,a}$.

Thm (MS 15): The protocol is secure against <u>any</u> adversary if $C > W_{G,a}$.

Best possible!

How much randomness (MS 15)



noise threshold vs.
of random bits per round

$$y = \frac{2.88(x - W_{G,a})^2}{|output| - 1}$$

Proof Techniques

A Mathematical Preliminary

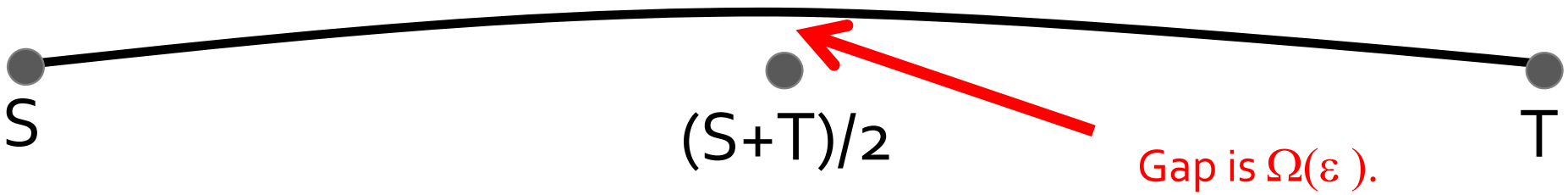
Consider the function $f(X) = \mathrm{Tr}\left[\ |X|^{1+\varepsilon}\right]$.

- If X is a density operator, f measures how deterministic X is. (Smaller = more random.)

- f is “almost” a norm on Hermitian operators.

A Mathematical Preliminary

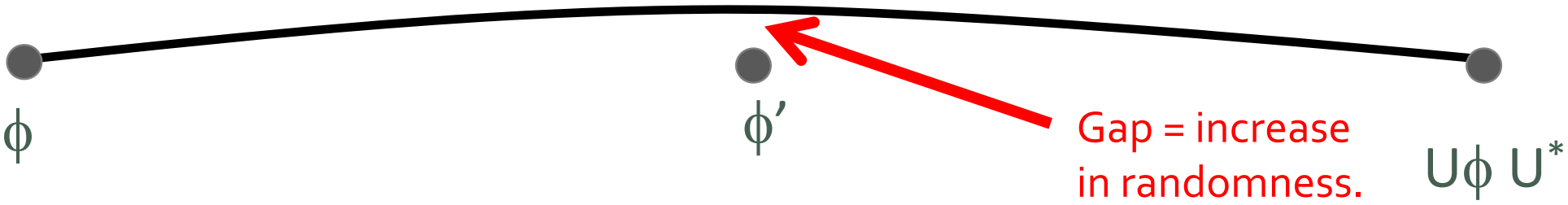
The function Tr [$|X|^{1+\varepsilon}$] is uniformly convex. [Ball+ 94]



A Mathematical Preliminary

Consequence [MS 15]: Suppose ϕ |–> ϕ' is the result of a binary measurement.

$$\phi' = \frac{\phi + U\phi U^*}{2}$$



The more **disturbance** caused by a measurement, the more **randomness** it adds.

Call this the $(1+\varepsilon)$–**uncertainty principle.**

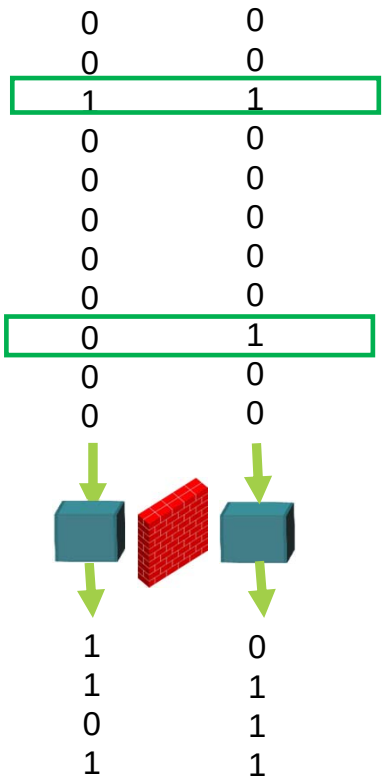
How do we prove security for this protocol?

Let G = nonlocal game, a = fixed input.

1. Run the device N times. During "game rounds," play G. Otherwise, just input a.

2. If the average score during game rounds was < C, abort.

3. Otherwise, apply randomness extractor.



How do we prove security

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1. Run the device N times. During "game rounds," play G. Otherwise, just input a.

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A starting point:

Suppose π is a function such that any device satisfies

$$H(\text{output} \mid \text{input} = a) \geq \pi(P(\text{win}))$$

Prop (easy)**:** In the **non-adversarial IID case,** the protocol produces at least $\pi(C)\,N$ extractable bits.

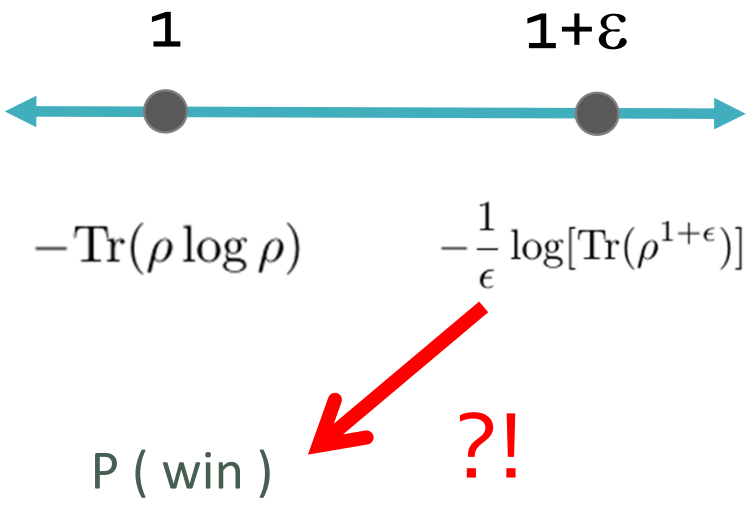
π = "simple rate curve"

What about non-IID?

Compare:

* **von Neumann** entropy (H)
* **Renyi** entropy ($H_{1+\varepsilon}$).

$H_{1+\varepsilon}$ proves extractable bits in the non-IID case!
But it's hard to relate to the winning probability.



What about non-IID?

Compare:

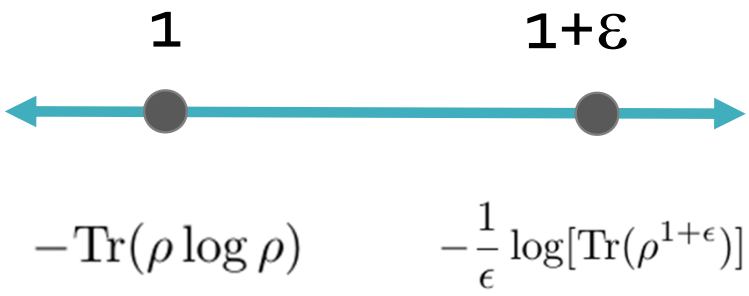
* **von Neumann** entropy (H)
* **Renyi** entropy ($H_{1+\varepsilon}$).

$H_{1+\varepsilon}$ proves extractable bits in the non-IID case!
But it's hard to relate to the winning probability.

Def: the (1+ε)**-winning probability** of a device is

$$\frac{\mathrm{Tr}[\rho_{win}^{1+\epsilon}]}{\mathrm{Tr}[\rho^{1+\epsilon}]}$$

where ρ = adversary's state.



What about non-IID?

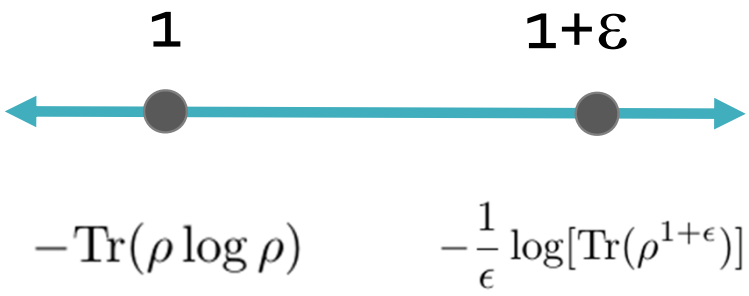
Def: π is a **strong rate curve** for the game G on input a if for all devices D,

$$H_{1+\varepsilon}\,(\text{output on input a} \mid \text{adversary})$$

is greater than or equal to

$$\pi\,(P_{1+\varepsilon}\,(\text{win})) + O_{\text{dev.-ind.}}(\varepsilon).$$

Thm [MS 15]: If π is a strong rate curve, then the spot-checking protocol produces $N \cdot \pi\,(C)$ extractable bits. (N = # of rounds, C = noise threshold.)



How do we prove strong rate curves?

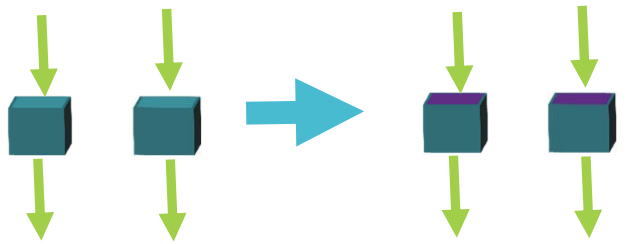
Want: High $P_{1+\varepsilon}$ (win) implies high $H_{1+\varepsilon}$.

Create a new device by pre-measuring w/ input a.

If this brings the score down significantly, then a significant amount of state disturbance has occurred. **(1+ε)–uncertainty principle** says that randomness was generated!

So if $P_{1+\varepsilon}$(win) is significantly larger than $W_{G,a}$, we have randomness.

Pre-apply the measurement for input a.

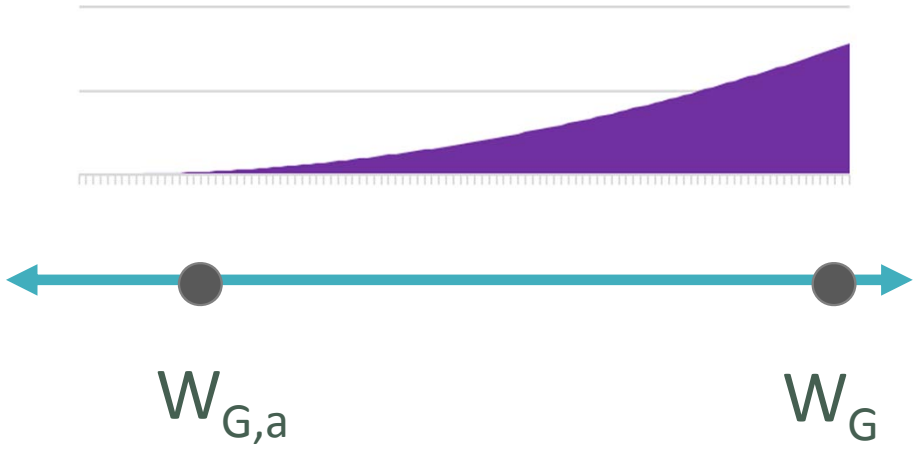


$P_{1+\varepsilon}$(win) vs. $P_{1+\varepsilon}$(win) <= $W_{G,a}$

The universal rate curves

Thm: For any (G,a) the function

$$y = \frac{2.88(x - W_{G,a})^2}{|output| - 1}$$



is a strong rate curve.

Noise tolerance for CHSH game: 10.3%!

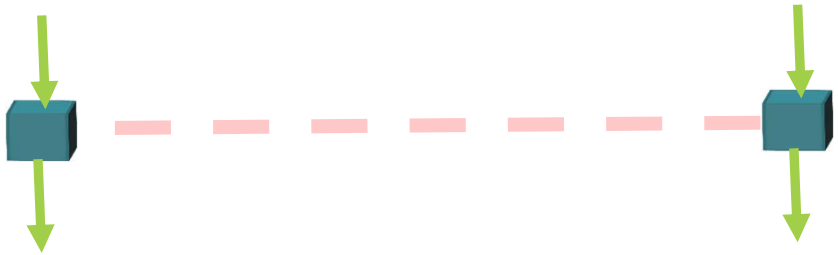
Addendum: DI-QKD

Device-independent Quantum Key Distribution

Our proof can be adapted to give another proof of DI-QKD.

1. Do step 1 of the spot-checking protocol. Communicate to check score.







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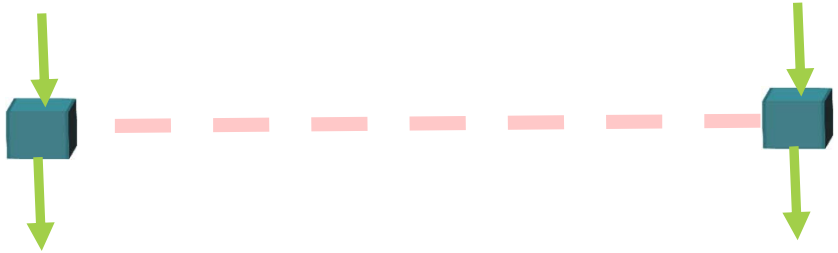
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Device-independent Quantum Key Distribution

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2. Have Alice make an optimal guess at Bob's bits using her bits.







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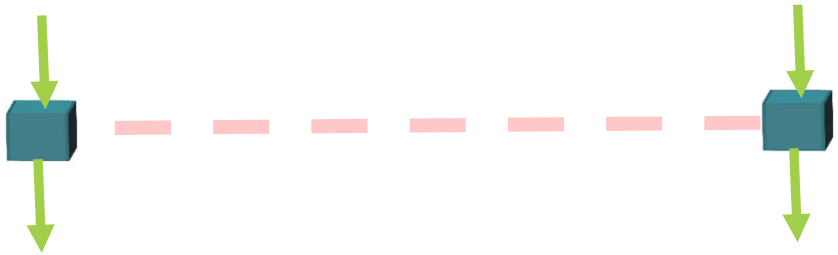
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1. Do step 1 of the spot-checking protocol. Communicate to check score.
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3. Perform information reconciliation.







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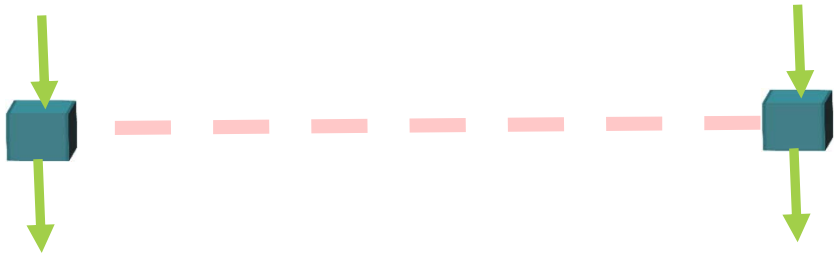
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1. Do step 1 of the spot-checking protocol. Communicate to check score.
2. Have Alice make an optimal guess at Bob's bits using her bits.
3. Perform information reconciliation.
4. Perform randomness extraction.

This works if step 1 generates more entropy than is lost at step 3.







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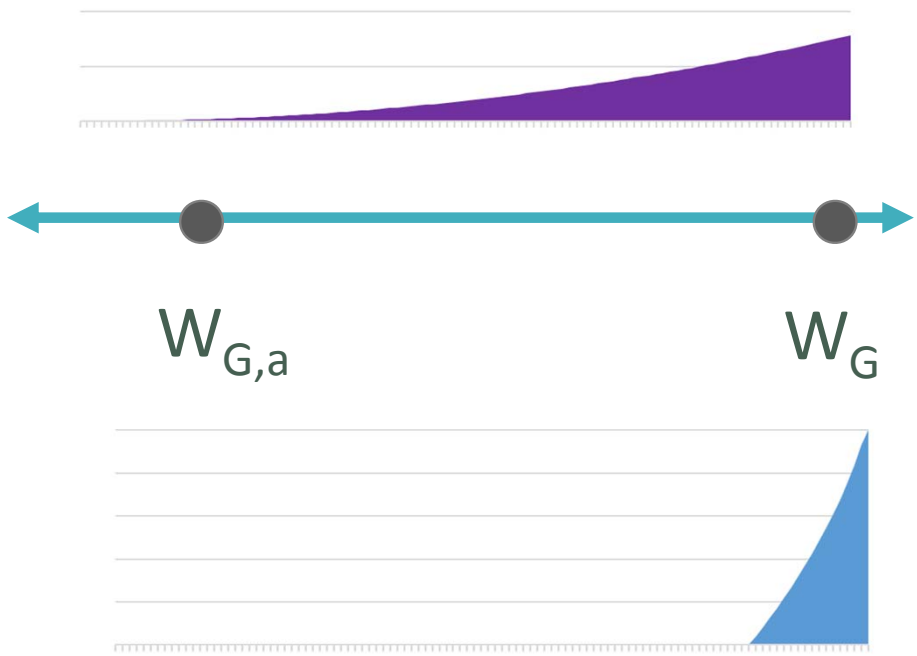
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Open Problems

Prove the best possible rate curves

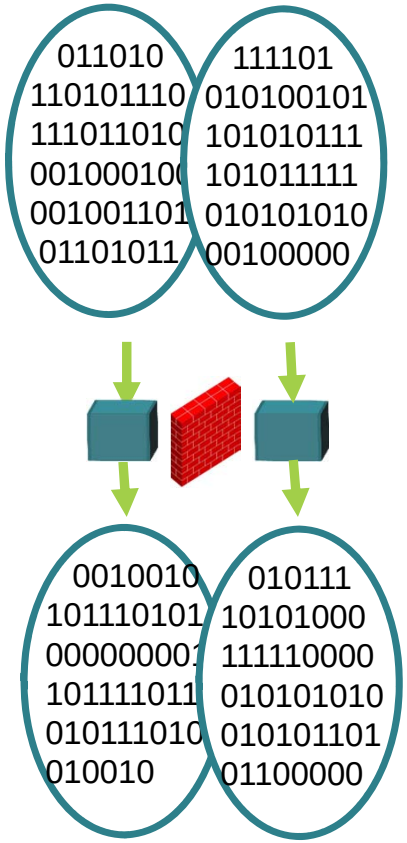
We have two families of rate curves, neither optimal. What are the best rate curves?

Can we match the classical-adversary rate curves?



Parallel randomness expansion

Give inputs to the boxes all at once. Can we still verify randomness?



Trustworthy Quantum Information

An International Workshop, June 28 – July 2, 2015, Ann Arbor, Michigan, USA



Conclusion

TYQI 2015 took place from 28th June to 2nd July. Click here to view the presentation slides at the event.



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With thanks to Zhu Cao, Dong-Ling Deng, Patrick Ion, Kihwan Kim, Christopher Portmann, Karl Winsor, Xiao Yuan, and Qi Zhao.